

?? Q. Write a PHP code for inserting, updating and deleting the Department table

1. Setup.sql

```
-- 1. Create the Database
CREATE DATABASE company_db;
USE company_db;

-- 2. Create the Department Table
CREATE TABLE department (
    dept_no VARCHAR(10) PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL,
    location VARCHAR(50)
);
```

2. insertionform.php

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Department Input Form</title>
    <style>
        label {
            display: inline-block;
            width: 120px;
            font-weight: bold;
        }
    </style>
</head>
<body>

    <fieldset>
        <legend>Department Information</legend>
        <form method="POST" action="dept_insert.php">

            <p>
                <label for="dept_no"> Department Number :</label>
                <input type="text" name="deptno" id="deptno" placeholder="e.g. D1">
            </p>

            <p>
                <label for="dept_name">Department Name :</label>
                <input type="text" name="deptname" id="deptname" placeholder="e.g. Sales">
            </p>

            <p>
                <label for="dept_location">Department Location :</label>
```

```

        <input type="text" name="deptlocation" id="deptlocation" placeholder="e.g. New
York">
    </p>

    <input type="submit" name="submit" value="Submit">
</form>
</fieldset>

</body>
</html>

```

3. connection.php

```

<?php
$servername = 'localhost';
$user = 'root';
$password = '';
$dbase = 'company_db';

$conn = mysqli_connect($servername, $user, $password, $dbase);

if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
?>

```

4. dept_insert.php

```

<?php
include 'connection.php';
if (isset($_POST['submit'])) {

    $dept_no = mysqli_real_escape_string($conn, $_POST['deptno']);
    $dept_name = mysqli_real_escape_string($conn, $_POST['deptname']);
    $location = mysqli_real_escape_string($conn, $_POST['deptlocation']);
    $sql = "INSERT INTO department (dept_no, dept_name, location)
        VALUES ('$dept_no', '$dept_name', '$location')";

    if (mysqli_query($conn, $sql)) {
        echo "New record entered successfully";
    } else {
        echo "Error: " . mysqli_error($conn);
    }
}
?>

```

OR

```

<?php
// Force error reporting on so we can see what's wrong
error_reporting(E_ALL);
ini_set('display_errors', 1);

echo "Step 1: Script started...<br>";

include 'connection.php';

if (isset($_POST['submit'])) {
    echo "Step 2: Submit button detected...<br>";

    $dep_no = mysqli_real_escape_string($conn, $_POST['deptno']);
    $dep_name = mysqli_real_escape_string($conn, $_POST['deptname']);
    $location = mysqli_real_escape_string($conn, $_POST['deptlocation']);

    $sql = "INSERT INTO department (dept_no, dept_name, location)
            VALUES ('$dep_no', '$dep_name', '$location')";

    if (mysqli_query($conn, $sql)) {
        echo "Step 3: Success! Record entered.";
    } else {
        echo "Step 3: Database Error: " . mysqli_error($conn);
    }
} else {
    echo "Step 2 FAILED: The script didn't see the 'submit' button. Check your HTML name
attribute.";
}
?>

```

5. Updating database: update_dept.php

```

<?php
include_once 'connection.php';

$msg = " ";

// 1. PROCESS THE UPDATE (If the form was submitted)
if (isset($_POST['update_btn'])) {
    $d_no = mysqli_real_escape_string($conn, $_POST['deptno']);
    $d_name = mysqli_real_escape_string($conn, $_POST['deptname']);
    $d_loc = mysqli_real_escape_string($conn, $_POST['deptlocation']);

    $sql = "UPDATE department
            SET dept_name = '$d_name',
                location = '$d_loc'
            WHERE dept_no = '$d_no'";

```

```

    if (mysqli_query($conn, $sql)) {
        $msg = "<p style='color: green;'>Record updated successfully!</p>";
    } else {
        $msg = "<p style='color: red;'>Error: " . mysqli_error($conn) . "</p>";
    }
}

```

// 2. FETCH DATA TO FILL THE FORM

// We need to know WHICH department to show.

// Usually, you pass this in the URL like: update_dept.php?deptno=D1

```
$id = isset($_GET['deptno']) ? $_GET['deptno'] : ($_POST['deptno'] ?? "");
```

```
$row = null;
```

```
if ($id != "") {
```

```
    $result = mysqli_query($conn, "SELECT * FROM department WHERE dept_no = '$id'");
```

```
    $row = mysqli_fetch_array($result);
```

```
}
```

```
?>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Update Department</title>
```

```
</head>
```

```
<body>
```

```
    <h2>Edit Department Details</h2>
```

```
    <?php echo $msg; // Show success or error message ?>
```

```
    <?php if ($row): ?>
```

```
        <form method="POST" action="update_dept.php">
```

```
            <p>
```

```
                <label>Department Number :</label><br>
```

```
                <!-- We use a hidden input to send the ID back to the PHP -->
```

```
                <input type="hidden" name="deptno" value="<?php echo $row['dept_no']; ?>">
```

```
                <strong><?php echo $row['dept_no']; ?></strong>
```

```
            </p>
```

```
            <p>
```

```
                <label>Department Name:</label><br>
```

```
                <input type="text" name="deptname" value="<?php echo $row['dept_name']; ?>">
```

```
            </p>
```

```
            <p>
```

```
                <label>Location:</label><br>
```

```

        <input type="text" name="deptlocation" value="<?php echo $row['location']; ?>">
    </p>

    <input type="submit" name="update_btn" value="Save Changes">
</form>
<?php else: ?>
    <p>No department selected to edit. Please provide a department number</p>
<?php endif; ?>

<hr>
<h3>Current Departments</h3>
<table border="1" cellpadding="10">
    <tr>
        <th>Dept No</th>
        <th>Name</th>
        <th>Location</th>
        <th>Action</th>
    </tr>
<?php
    $all_depts = mysqli_query($conn, "SELECT * FROM department");
    while($row_list = mysqli_fetch_array($all_depts)) {
        echo "<tr>";
        echo "<td>" . $row_list['dept_no'] . "</td>";
        echo "<td>" . $row_list['dept_name'] . "</td>";
        echo "<td>" . $row_list['location'] . "</td>";
        echo "<td><a href='update_dept.php?deptno=" . $row_list['dept_no'] . "'>Edit</a></td>";
        echo "</tr>";
    }
    ?>
</table>
</body>
</html>

```

6. Deleting entries : delete_dept.php

```

<?php
include_once 'connection.php';

$msg = "";

// 1. PROCESS THE DELETE (If 'id' is sent via the URL)
if (isset($_GET['deptno'])) {
    $id = mysqli_real_escape_string($conn, $_GET['deptno']);

    // SQL to delete a specific record
    $sql = "DELETE FROM department WHERE dept_no = '$id'";

    if (mysqli_query($conn, $sql)) {
        $msg = "<p style='color: green;'>Department $id deleted successfully!</p>";
    }
}

```

```

    } else {
        $msg = "<p style='color: red;'>Error deleting record: " . mysqli_error($conn) . "</p>";
    }
}
?>

```

```

<!DOCTYPE html>

```

```

<html>

```

```

<head>

```

```

    <title>Delete Department</title>

```

```

</head>

```

```

<body>

```

```

    <h2>Delete Department</h2>

```

```

    <?php echo $msg; ?>

```

```

    <hr>

```

```

    <h3>Select a Department to Delete</h3>

```

```

    <table border="1" cellpadding="10">

```

```

        <tr>

```

```

            <th>Dept No</th>

```

```

            <th>Name</th>

```

```

            <th>Action</th>

```

```

        </tr>

```

```

        <?php

```

```

            $result = mysqli_query($conn, "SELECT * FROM department");

```

```

            while($row = mysqli_fetch_array($result)) {

```

```

                echo "<tr>";

```

```

                echo "<td>" . $row['dept_no'] . "</td>";

```

```

                echo "<td>" . $row['dept_name'] . "</td>";

```

```

                echo "<td><a href='delete_dept.php?deptno=" . $row['dept_no'] . "'

```

```

                    onclick=\"return confirm('Are you sure you want to delete
this?');\">Delete</a></td>";

```

```

                echo "</tr>";

```

```

            }

```

```

        ?>

```

```

    </table>

```

```

    <br>

```

```

    <a href="update_dept.php">Back to Update Page</a>

```

```

</body>

```

```

</html>

```